

PAPER The REBCO High-Field Superconductor Basis: Conductor, Winding, and the Field Envelope for the Breeder Centrepost and Burner Plug

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Abstract

The enabling technology of compact fusion is the high-field rare-earth barium copper oxide (REBCO) magnet, and the binding engineering limit of such a magnet is not the coated conductor's critical current but the irreversible strain the winding may carry under Lorentz load. We present a design-and-analysis study of the REBCO *superconductor basis* shared by the two dissimilar machines of the Kronos platform: the centrepost of a compact spherical-tokamak breeder (Hyperion; peak field 16.84 T, $B_0 = 8$ T on axis, $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW) and the high-field end-plug of a D-³He tandem-mirror burner (design field 26.49 T, $Q_E = 1.318$). We separate cleanly what the *conductor* enables from what a *coil* can realise. REBCO's upper critical field exceeds 100 T, so a 35 T conductor operating point is roughly one-third of the material limit; whether a coil reaches its field is a bore-and-stress question, not a conductor-current question. We set out the governing mechanics as explicit equations — the Lorentz body force, the winding-elasticity boundary-value problem, its closed-form Lamé reduction for a reinforced cylinder, the critical-surface current-sharing margin, the adiabatic quench integral, and the neutron-fluence life law — and reduce them to practice under one bore-resolved finite-element method benchmarked against the closed form. Both magnets qualify structurally within demonstrated reinforcement practice. The breeder centrepost is stress-comfortable (load fraction $I_{\text{op}}/I_c \approx 0.27$, temperature margin $\Delta T \approx 49.5$ K); its binding limit is the neutron lifetime of the winding (18.85 dpa fpy⁻¹, a 0.58-full-power-year cartridge life at full inboard shielding), not mechanical stress. The burner plug, at its required bore, carries a magnetic pressure of 115/279/358 MPa at 17/26.49/30 T and meets a sub-metre reinforcement against a 667 MPa allowable, the finite-element stress ratio passing at 0.66; its temperature margin is $\Delta T \approx 37.1$ K at load fraction 0.43. A graded per-turn pre-load flattens the Lorentz strain gradient and cuts the peak REBCO hoop strain by 52.4% ($0.595\% \rightarrow 0.283\%$ at 26.49 T), holding the winding under the 0.40% irreversible limit even with thermal and vertical loads superposed. Every performance number is a design target, a model output, or an inference from public REBCO literature and is a frozen anchor in the Kronos de-risking register; nothing has been fabricated or measured.

Keywords: REBCO coated conductor; high-field magnet; winding hoop strain; irreversible-strain limit; Lamé mechanics; no-insulation winding; quench protection; neutron irradiation; compact fusion

Nomenclature

Symbol	Definition	Symbol	Definition
B_{c2}	upper critical field (> 100 T)	p_{mag}	magnetic pressure $B^2/2\mu_0$
σ_θ, σ_r	hoop, radial stress	ε_θ	hoop strain
ε_{irr}	irreversible strain limit (0.40%)	E_w, ν	winding modulus, Poisson ratio
a, b_w, b	bore, pack-OD, reinforced-OD radius	t_{req}	required reinforcement thickness
$J, \mathbf{f}$	current density, body force	$u(r)$	radial displacement
I_c, I_{op}	critical, operating current	ΔT	current-sharing temperature margin
$T_{\text{cs}}, T_{\text{op}}$	current-sharing, operating temperature	v_{NZP}	normal-zone propagation velocity
$\Phi, \dot{\Phi}$	fast fluence, fluence rate	τ_{life}	centrepost cartridge life
NI/MI	no-insulation / metal-insulation	σ_{allow}	reinforced allowable (667 MPa)

1 Introduction

1.1 REBCO as the enabling material of compact fusion

Compact fusion is enabled by rare-earth barium copper oxide (REBCO) coated conductors because REBCO carries transport current at magnetic fields no low-temperature superconductor reaches [14, 6]. The fusion power density of a magnetically confined plasma scales steeply with the field on the plasma, $P/V \propto \beta^2 B^4$, so a factor-of-two increase in achievable field is a factor-of-sixteen lever on power density and the central reason a device can be made small [12, 11]. Two independent programmes have now shown that REBCO delivers the field: a laboratory solenoid has reached 45.5 T of direct-current field [3], and a full-scale toroidal-field model coil for a compact tokamak has demonstrated ~ 20 T on the conductor at reactor scale [11, 7]. High current density at high field is therefore no longer the open question; the open question is the *magnet*.

The Kronos platform uses two distinct REBCO magnets, and the paper’s first premise is that they must be treated as the distinct devices they are (register ruling KX-L1-A15). The spherical-tokamak breeder, Hyperion, is organised around a single-turn-equivalent *centrepost* magnet at a peak field of 16.84 T, running the plasma current $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind it; the D-³He tandem-mirror burner is organised around a much higher-field *end-plug* magnet whose design field is 26.49 T [28, 29, 30]. These are not the same coil at two operating points — they differ in field, bore, duty and radiation environment — and conflating the centrepost field with the plug field is a category error that the analysis below is careful never to make.

1.2 Strain-first: the binding limit is irreversible winding strain

The field a magnet can hold is not set by the conductor’s critical current; it is set by the strain the winding can carry before the coated conductor degrades *irreversibly* [1]. A REBCO tape is a brittle 1 μ m-thick oxide film epitaxially grown on a metal substrate; its critical current is reversible under modest tensile strain but falls irreversibly once a threshold near $\varepsilon_{\text{irr}} \approx 0.4\%$ is exceeded [1, 2]. Because the Lorentz body force in a high-field winding produces exactly a hoop *tension*, the irreversible-strain limit — not the critical current, and not even the yield of the structural alloy — is the true design ceiling of a high-field REBCO magnet. We therefore adopt a *strain-first* acceptance criterion throughout: a coil qualifies when its peak winding hoop strain stays below ε_{irr} under all superposed loads, with reinforcement added until it does.

1.3 Contribution and organisation

This paper is the superconductor-basis foundation of the Kronos 2026 design series. Its contribution is fourfold. (i) It states the governing mechanics of a high-field REBCO winding as a closed set of equations (§2): the body force, the winding-elasticity boundary-value problem, the closed-form Lamé reduction, the critical-surface margin, the adiabatic quench integral, and the neutron-fluence life law. (ii) It gives the GPU-accelerated computational methodology — the real codes used, and the closed-form benchmark that validates them (§3). (iii) It presents the results (§4): the field envelope, the bore-resolved mechanics, the magnetic-pressure and reinforcement budgets, the strain-first envelope with graded pre-load, the temperature-margin load line of the two magnets, the neutron-lifetime and irradiation-tolerance results, and the fused quench-precursor detection, across eleven figures and four tables. (iv) It compares each result to the published REBCO, high-field-magnet and quench-detection literature (§5), and states honestly what has not been built (§6). Every quantitative claim traces to a frozen serial in the Kronos Physics De-Risking Register [28]; the serials (e.g. BU-L1-A2, SC-BR-4) are given inline so each number is auditable.

2 Governing equations and the formal problem

2.1 Lorentz body force and magnetic pressure

A winding carrying current density $\mathbf{J}$ in field $\mathbf{B}$ experiences the Lorentz body force

$$\mathbf{f} = \mathbf{J} \times \mathbf{B}. \quad (1)$$

For a solenoidal winding the axial current interacts with the radial gradient of the axial field to produce a net outward radial force; integrated across the winding this is equivalent to an outward *magnetic pressure* acting on the bore,

$$p_{\text{mag}} = \frac{B^2}{2\mu_0}, \quad (2)$$

the magnetostatic Maxwell stress of a field B . Equation (2) is the load that every subsequent stress follows: at the plug operating fields it evaluates to

$$p_{\text{mag}} = 115, 279, 358 \text{ MPa} \quad \text{at} \quad B = 17, 26.49, 30 \text{ T} \quad (3)$$

(register BU-L1-A2), the middle value being the plug design field. The quadratic dependence $p_{\text{mag}} \propto B^2$ is the single most important scaling in the paper: it is why the plug coil, at 26.49 T, carries roughly 2.5× the bore pressure of the centrepost at 16.84 T, and why a small increase in field demands a disproportionate increase in reinforcement.

2.2 Winding elasticity: the boundary-value problem

Resolving the finite wall requires the elasticity of a thick, orthotropic cylinder under a distributed body force. Working in axisymmetric plane strain with radial displacement $u(r)$, the equilibrium of a differential winding element is

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} + f_r(r) = 0, \quad (4)$$

with the Lorentz radial body force $f_r(r) = J_\phi(r) B_z(r)$ from equation (1). The winding pack is a composite of REBCO tape, co-wound stabiliser and reinforcement; we treat it as a homogenised orthotropic continuum with the strain–displacement relations

$$\varepsilon_r = \frac{du}{dr}, \quad \varepsilon_\theta = \frac{u}{r}, \quad (5)$$

and the plane-strain constitutive law

$$\begin{aligned} \sigma_r &= \frac{E_w}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_r + \nu\varepsilon_\theta], \\ \sigma_\theta &= \frac{E_w}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_\theta + \nu\varepsilon_r], \end{aligned} \quad (6)$$

where E_w is the effective winding-pack modulus and ν its Poisson ratio. Substituting (5)–(6) into (4) yields the Navier–Lamé ordinary differential equation for the displacement,

$$\frac{d^2u}{dr^2} + \frac{1}{r} \frac{du}{dr} - \frac{u}{r^2} = - \frac{(1+\nu)(1-2\nu)}{E_w(1-\nu)} f_r(r), \quad (7)$$

to be solved on the winding annulus $a \leq r \leq b$ with traction-free inner and outer boundaries, $\sigma_r(a) = \sigma_r(b) = 0$ (the bore and the outer overwrap carry no external pressure). Equations (4)–(7) are the boundary-value problem the finite-element solver integrates for the true, graded-property winding.

2.3 Closed-form Lamé reduction and the reinforcement budget

When the body force is idealised as an equivalent internal pressure p_{mag} acting on the bore of a uniform-property reinforced cylinder of inner radius a and outer radius b , equation (7) admits the classical Lamé solution [25, 26]. The hoop stress is

$$\sigma_\theta(r) = \frac{a^2 p_{\text{mag}}}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right), \quad (8)$$

which is maximised at the inner bore $r = a$,

$$\sigma_\theta^{\text{max}} = \sigma_\theta(a) = p_{\text{mag}} \frac{b^2 + a^2}{b^2 - a^2}. \quad (9)$$

Equation (9) sets the reinforcement budget directly. Requiring $\sigma_\theta^{\text{max}} \leq \sigma_{\text{allow}}$ and solving for the outer radius gives the required reinforced wall,

$$t_{\text{req}} = b - a = a \left[\sqrt{\frac{\sigma_{\text{allow}} + p_{\text{mag}}}{\sigma_{\text{allow}} - p_{\text{mag}}}} - 1 \right], \quad (10)$$

finite only while $p_{\text{mag}} < \sigma_{\text{allow}}$ — the condition that a reinforced monolith can hold the field at all. With the reinforced allowable $\sigma_{\text{allow}} = 667 \text{ MPa}$ (a demonstrated high-strength structural value for a co-wound/overwrapped winding pack), equation (10) evaluates to $t_{\text{req}} = 0.16, 0.48, 0.71 \text{ m}$ at the three plug fields of (3) — sub-metre reinforcement across the range (register BU-L1-A2).

2.4 The strain-first acceptance criterion

The strain-first ceiling replaces a bare stress allowable with the coated conductor's irreversible-strain limit. With the peak hoop stress of equation (9) mapped through the winding modulus, the acceptance criterion is

$$\varepsilon_{\theta}^{\max} = \frac{\sigma_{\theta}^{\max}}{E_w} \leq \varepsilon_{\text{irr}} = 0.40\%, \quad (11)$$

the limit at which the REBCO film degrades irreversibly [1]. Reinforcement is added — raising the effective b and E_w in (8)–(11) — until (11) holds with margin. A crucial refinement is that the *peak* strain, not the average, governs, and the peak sits at the bore (equation 8); a winding that meets a section-average strain can still fail at the innermost turn. This is the physical motivation for the graded per-turn pre-load of §4.5: by placing a controlled compressive pre-strain that is largest at the bore and tapers outward, one flattens the Lorentz-driven gradient and lowers the governing peak.

2.5 Critical surface and current-sharing temperature margin

Whether a turn is superconducting at all is governed by the REBCO critical surface $I_c(B, T, \theta)$, the locus at which flux flow begins. For an operating current I_{op} the *current-sharing temperature* T_{cs} is the temperature at which I_c falls to I_{op} ,

$$I_c(B, T_{cs}, \theta) = I_{\text{op}}, \quad (12)$$

and the protective observable is the current-sharing temperature margin

$$\Delta T = T_{cs} - T_{\text{op}}, \quad (13)$$

the thermal headroom before the turn shares current with its stabiliser. The load fraction $I_{\text{op}}/I_c(B, T_{\text{op}})$ measures how hard the conductor is driven. A convenient engineering parametrisation of the field and temperature dependence, in the tradition of critical-state fits [16, 15], is the separable power law

$$I_c(B, T, \theta) = I_{c0} g(\theta) \left(\frac{B}{B_0} \right)^{-\alpha} \left(1 - \frac{T}{T_c(B)} \right)^{\beta}, \quad (14)$$

with anisotropy factor $g(\theta)$, exponents α, β fitted to tape data, and $T_c(B)$ the field-depressed transition temperature. Evaluating (12)–(14) on each magnet's own critical surface (register KX-L1-A15) gives the two working points of §4.6: the centrepost at $\Delta T \approx 49.5$ K, load fraction ≈ 0.27 , and the plug at $\Delta T \approx 37.1$ K, load fraction ≈ 0.43 . The large margins are a direct consequence of the field envelope of §4.1: both magnets run far below B_{c2} , so both retain tens of kelvin of thermal headroom.

2.6 No-insulation winding and the quench integral

The Kronos magnets are wound from no-insulation (NI) or metal-insulation (MI) REBCO, in which adjacent turns are in direct (or resistively bridged) electrical contact [5, 4]. NI winding gives self-protection and mechanical robustness but a slow charging response set by the characteristic turn-to-turn time constant [27]

$$\tau_{\text{NI}} \sim \frac{L}{R_{ct}/N_t}, \quad (15)$$

with L the coil inductance, R_{ct} the areal contact resistance and N_t the turn count; MI raises R_{ct} to trade ramp delay against quench protection. The protection problem itself is that NI REBCO has a very slow normal-zone propagation velocity v_{NZP} , of order millimetres per second [19], so a hot spot can reach a damaging temperature before it spreads. The hot-spot temperature is bounded by the adiabatic *MIITs* integral,

$$\int_0^{t_{\text{dump}}} J(t)^2 dt = \int_{T_{\text{op}}}^{T_{\text{max}}} \frac{C(T)}{\rho(T)} dT \equiv Z(T_{\text{max}}), \quad (16)$$

where $C(T)$ and $\rho(T)$ are the volumetric heat capacity and resistivity of the composite and $Z(T_{\text{max}})$ is a material property. For a constant stabiliser current density J , equation (16) gives the allowable detect-plus-dump time $t_{\text{dump}} = Z(T_{\text{max}})/J^2$: at a stabiliser density of 100 A mm^{-2} the winding tolerates ~ 4 s before exceeding 150 K, shrinking to ~ 1.9 s at 200 A mm^{-2} (register HX-10). This finite, seconds-scale budget is what makes early quench precursor detection (§4.9) valuable rather than merely nice to have.

2.7 Neutron damage and the winding life law

On the breeder centrepost the winding is a consumable under 14 MeV neutron load. The displacement-damage rate follows the Norgett–Robinson–Torrens (NRT) standard [17] from the damage-energy cross-section, giving $18.85 \text{ dpa fpy}^{-1}$ at the centrepost first wall (register BR-L1-A13). The irreversible degradation of REBCO I_c under fast neutron fluence sets a break-even fluence Φ_{be} beyond which the tape must be retired [9]; the cartridge life is then

$$\tau_{\text{life}} = \frac{\Phi_{\text{be}}}{\dot{\Phi}(t_{\text{sh}})}, \quad (17)$$

where the fast-fluence rate $\dot{\Phi}$ is a steep function of the inboard shield thickness t_{sh} . A tungsten-carbide inboard shield of 0–45 cm cuts $\dot{\Phi}$ from 2.9×10^{25} to $1.7 \times 10^{23} \text{ m}^2/\text{fpy}$, moving τ_{life} from an inadmissible 0.003 to 0.58 fpy (register BR-L1-A13b). Equation (17) is why the centrepost is designed as a scheduled-replacement cartridge and why its binding limit is neutron lifetime, not stress. The burner plug, sited behind a low-neutron D– ^3He plasma, does not share this constraint.

3 Computational methodology (GPU-accelerated HPC)

Every headline number was produced by an explicit code, and the mechanics were cross-checked against the closed form of §2. The heavy transport and materials campaigns ran on the NVIDIA GPU HPC campaign (a mixed A100/H100/A10 fleet); the analytic mechanics, closed-form benchmarks and signal-analysis reductions ran on the in-house HPC node so that the fast checks never contended with the fleet. No result depends on proprietary data, and no economic quantity enters the analysis.

Winding mechanics. The bore-resolved hoop-stress profile (figure 3) was integrated with a finite-element solution of equations (4)–(7) for the graded-property winding (`sfepy`; register SC-BR-4, HX-04, HX-09) and independently with the closed-form Lamé expression (8). Three-dimensional structural verification on the as-drawn CAD used the `CalculiX` (`ccx`) solver (register P3-FEA-01 for the centrepost, P3-FEA-10 for the plug). The analytic Lamé envelope (register HX-11) reproduces the finite-element magnetic-pressure anchors of equation (3) node-for-node.

Neutronics. Displacement damage, fast fluence and the shield trade of equation (17) were computed with `OpenMC` Monte-Carlo transport and its damage-energy tally over ENDF/B-VIII.0 data [18] (register BR-L1-A13, BR-L1-A13b, P3-NEUT), with the NRT rule of [17].

Irradiation tolerance of the conductor. The break-even fluence and the un-doped-tape rule (figure 10) rest on molecular-dynamics displacement cascades in YBCO (`LAMMPS`, EAM potentials; register BR-L2-A36) and on first-principles chain-oxygen-vacancy formation energies computed with DFT+U (register companion, [36]); the NRT rulers set the dpa-to-fluence conversion.

Quench and sensing. The adiabatic MIITs budget of equation (16) was evaluated with a numerical hot-spot integrator (register HX-10). The fused strain-rate-plus-magnetization precursor ROC (figure 11) was computed by a signal-analysis and machine-learning reduction over a frozen 2×10^4 -trial synthetic set (`scikit-learn`; register KX-L1-A20), and the in-winding nitrogen-vacancy (NV) diamond magnetometer sensitivity by an analytic shot-noise model (`numpy/scipy`; register KX-L1-A22).

Verification. The mechanics are accepted only against the closed form: the finite-element hoop-stress profile of figure 3 tracks the Lamé solution across the winding with the peak at the inner bore in both, the finite-element peak sitting a few percent below the uniform-property closed form because the graded reinforcement is resolved. The three-dimensional CAD FEA closes the same result at higher fidelity, returning a plug stress ratio of 0.66 against the allowable (register BU-L1-A2, P3-FEA-10).

4 Results

4.1 The field envelope: what the conductor enables

REBCO’s upper critical field exceeds 100 T, so every Kronos operating point is a fraction of the material limit (figure 1). The breeder centrepost peak field of 16.84 T is $0.17 B_{c2}$; the burner plug design field of 26.49 T is $0.26 B_{c2}$; the plug qualification bound of 30 T is $0.30 B_{c2}$; and even the 35 T conductor-capability point — a claim about the tape, not a coil record — is only about one-third of the limit. The envelope makes the paper’s central separation concrete: the conductor is nowhere near its field ceiling, so the binding limit of each magnet is set elsewhere — by winding strain and neutron life, both analysed below — and not by the superconductor running out of critical current.

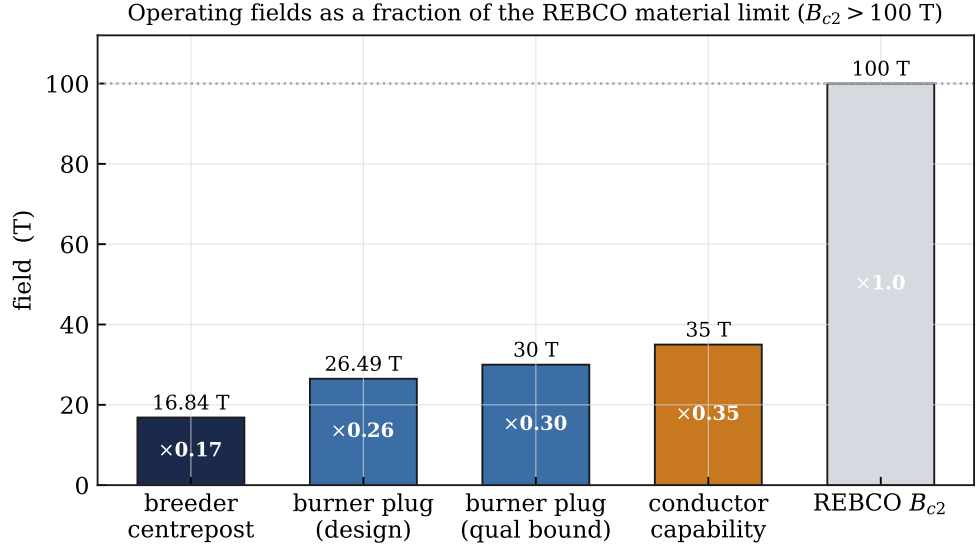


Figure 1. Operating fields as a fraction of the REBCO material limit. The breeder centrepost (16.84 T), burner plug design (26.49 T) and qualification (30 T) fields, and the 35 T conductor-capability point all sit well below $B_{c2} > 100$ T. Data: register field anchors.

4.2 The conductor architecture

The coated conductor is a multilayer stack (figure 2): the REBCO film on its buffered Hastelloy substrate, capped with silver, then a copper stabiliser and the reinforcement. The Kronos architecture layers on this three ideas grounded in prior art [2, 7]: a vanadium-oxide (V_2O_3) metal-insulator-transition “smart” interlayer that raises turn-to-turn resistance on heating and so mediates the NI/MI trade of equation (15); a graphene-reinforced aluminium bypass stabiliser; and a work-hardened reinforcement that carries the hoop load. Our contribution to the conductor is analysis, not invention, and we report honestly (as in the register) that one element — the aluminium bypass at the operating field — is unsupported by data and that a specific reinforcement-as-epitaxial-substrate route is infeasible; these are recorded rather than hidden.

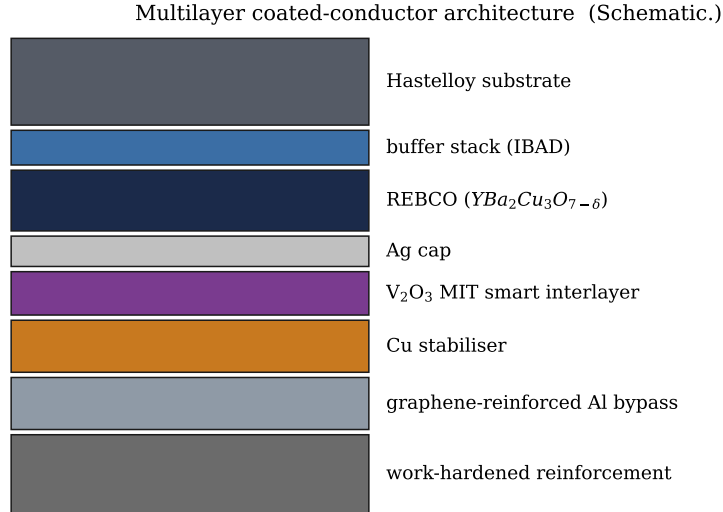


Figure 2. (Schematic.) Multilayer coated-conductor architecture: the REBCO film on a buffered Hastelloy substrate with silver cap and copper stabiliser, augmented by a V_2O_3 metal-insulator smart interlayer, a graphene-reinforced aluminium bypass and a work-hardened reinforcement. Layer order is indicative; no thicknesses to scale.

4.3 Bore-resolved winding mechanics

Figure 3 shows the bore-resolved hoop-stress profile from the finite-element integration of equations (4)–(7) against the closed-form Lamé solution of equation (8). Both peak at the inner

bore, as equation (9) requires; the finite-element peak of 625 MPa sits about five percent below the uniform-property closed form (657 MPa), the reduction reflecting the graded reinforcement the finite-element model resolves and the closed form does not. This agreement in shape and peak location is what licenses using the tractable closed form for the reinforcement budget while retaining the finite-element model for the governing peak. The reinforced thick-wall geometry the mechanics assume is sketched in figure 4.

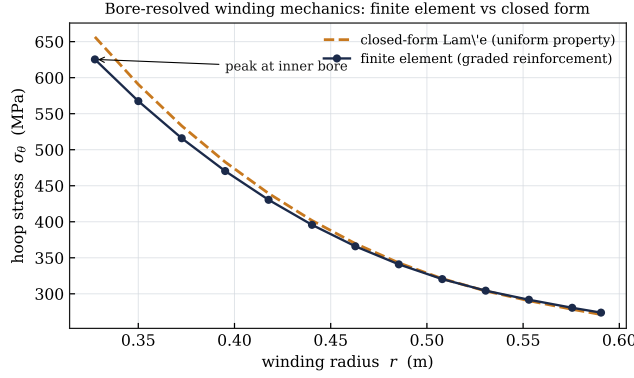


Figure 3. Bore-resolved winding mechanics: finite-element (graded reinforcement) versus closed-form Lamé (uniform property). Both peak at the inner bore; the finite-element peak sits a few percent lower where the grading is resolved. Data: register coil-FEA / SC-BR-4 nodes.

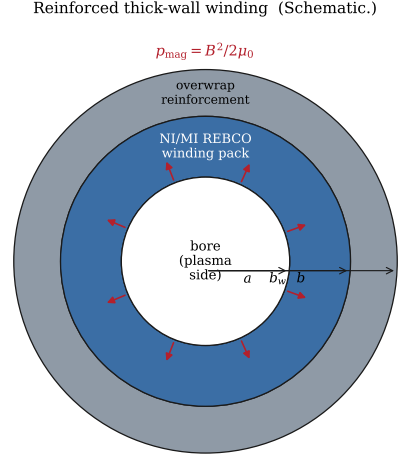


Figure 4. (Schematic.) Reinforced thick-wall winding: bore a , winding-pack outer radius b_w , reinforced outer radius b , under the outward magnetic pressure p_{mag} of equation (2).

4.4 Magnetic pressure and the reinforcement budget

The plug coil carries the heaviest bore load. Figure 5 plots the magnetic pressure of equation (3) against the reinforced allowable: 115/279/358 MPa at 17/26.49/30 T, all below the 667 MPa allowable, so a reinforced monolith can hold the field at every point (the condition $p_{\text{mag}} < \sigma_{\text{allow}}$ of equation 10 is met with room to spare). The corresponding required reinforcement thickness from equation (10) is plotted in figure 6: 0.16/0.48/0.71 m, sub-metre across the field range and within demonstrated overwrap practice. The quadratic growth of p_{mag} with field is visible directly as the steep rise of t_{req} from 17 T to 30 T.

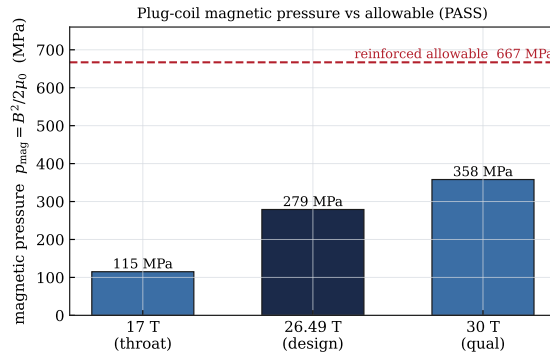


Figure 5. Plug-coil magnetic pressure $p_{\text{mag}} = B^2/2\mu_0$ at the three plug fields against the 667 MPa reinforced allowable (register BU-L1-A2). All three points pass.

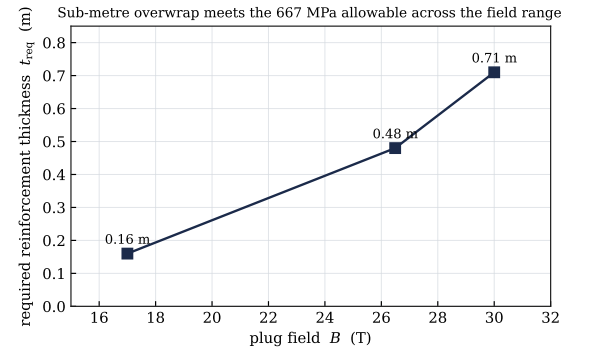


Figure 6. Required reinforcement thickness t_{req} from the Lamé budget of equation (10): sub-metre overwrap meets the allowable across 17–30 T (register BU-L1-A2).

4.5 The strain-first envelope and graded per-turn pre-load

The genuinely new engineering element is a graded, per-turn pre-load that flattens the Lorentz strain gradient of equation (8). Figure 7 plots the peak REBCO hoop strain against field for three cases, scaled from the frozen finite-element anchors through the $p_{\text{mag}} \propto B^2$ law. Without pre-load

the peak strain reaches 0.595% at 26.49 T — above the 0.40% irreversible limit. A graded per-turn pre-load that places the largest compressive pre-strain at the bore and tapers it outward cuts the peak by 52.4% to 0.283% and flattens the profile roughly fourfold (register SC-BR-4). With thermal contraction and vertical load superposed, the peak rises to 0.365% at the design field — still under the limit — and closes to the limit only at the 30 T qualification bound (register HX-09). On the as-drawn CAD the same graded pre-load returns a peak hoop strain of 0.151% on the centrepost (2.65 $\times$ margin, jacket von Mises 579 MPa, 1.38 $\times$) and 2.85–3.54 $\times$ margin on the plug (register P3-FEA-01, P3-FEA-10; the plug’s three-dimensional structural FEA is developed in a companion paper [38]). The strain-first criterion of equation (11) is therefore met with margin at both magnets, with the pre-load method — not a new conductor material — doing the work.

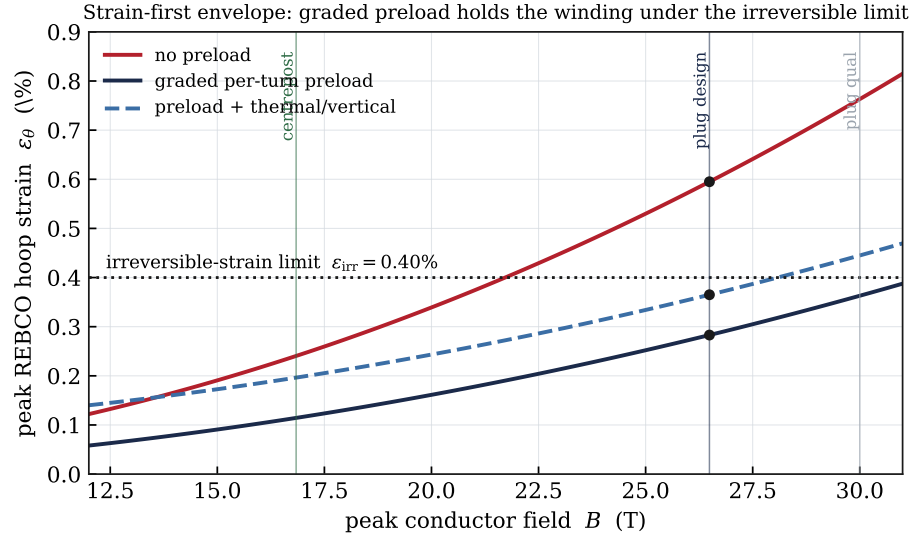


Figure 7. Strain-first envelope: peak REBCO hoop strain versus field for no pre-load, graded per-turn pre-load, and pre-load with thermal/vertical loads superposed, against the 0.40% irreversible limit. Curves are derived from the frozen finite-element anchors (register SC-BR-4, HX-09; markers) through the $p_{\text{mag}} \propto B^2$ Lorentz scaling, with the thermal/vertical contribution modelled as a field-independent offset.

4.6 The temperature-margin load line of the two magnets

Figure 8 places the two distinct magnets on the load-fraction/temperature-margin plane of equations (12)–(13). The breeder centrepost, at 16.84 T, runs at a load fraction $I_{\text{op}}/I_c \approx 0.27$ with a current-sharing margin $\Delta T \approx 49.5$ K; the burner plug, at the higher 26.49 T, runs harder at load fraction ≈ 0.43 with $\Delta T \approx 37.1$ K (register KX-L1-A15). Both retain tens of kelvin of thermal headroom, a direct consequence of operating far below B_{c2} (figure 1). Crucially, the two points lie on *different* critical surfaces: the twin that models them — and the protection that guards them — treats the centrepost field and the plug field as the distinct quantities they are, never conflating 16.84 T with 26.49 T.

4.7 Neutron lifetime of the breeder centrepost

Figure 9 shows the consumable-centrepost life law of equation (17). At the centrepost damage rate of 18.85 dpa fpy $^{-1}$, the fast-fluence rate — and hence the cartridge life — is a steep function of the inboard tungsten-carbide shield: a bare centrepost survives only 0.003 fpy, a 30 cm shield extends it to 0.221 fpy, and full (45 cm) shielding reaches 0.58 fpy, with the fast fluence falling two orders of magnitude from 2.9×10^{25} to 1.7×10^{23} m 2 /fpy (register BR-L1-A13b). The centrepost is therefore designed as a scheduled-replacement cartridge, breeding from the outboard side while the inboard winding is consumed; the trade of coverage against protection is the subject of a companion paper [33, 34].

4.8 Irradiation tolerance and the un-doped-tape rule

The break-even fluence Φ_{be} that enters equation (17) is itself a conductor-design result. Figure 10 compares the break-even fast fluence of an un-doped REBCO tape, 5.64×10^{22} m $^{-2}$, with that of a tape carrying engineered artificial pinning centres, 3.82×10^{22} m $^{-2}$ (register BR-L2-A36, HX-22). The un-doped tape survives 1.48 $\times$ more fluence: the pinning defects that raise in-field I_c are the

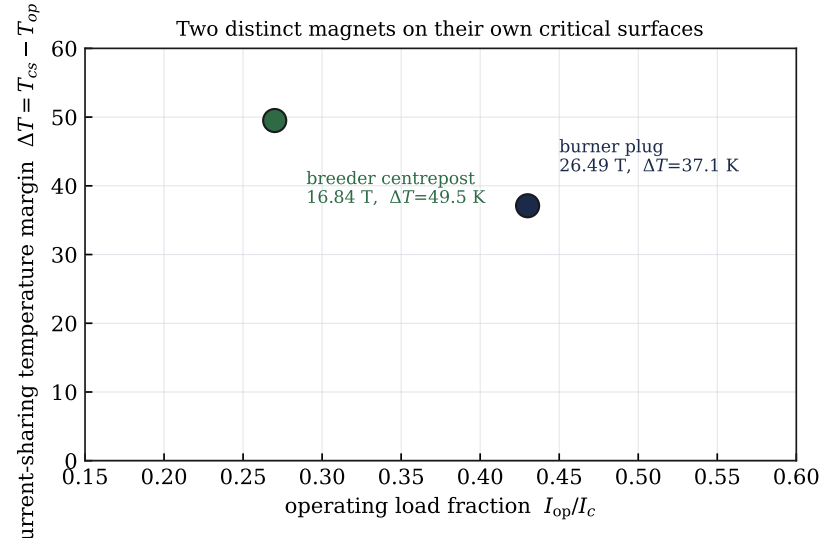


Figure 8. Current-sharing temperature margin against operating load fraction for the two distinct magnets (register KX-L1-A15): breeder centrepost $\Delta T \approx 49.5$ K at 16.84 T, load fraction 0.27; burner plug $\Delta T \approx 37.1$ K at 26.49 T, load fraction 0.43.

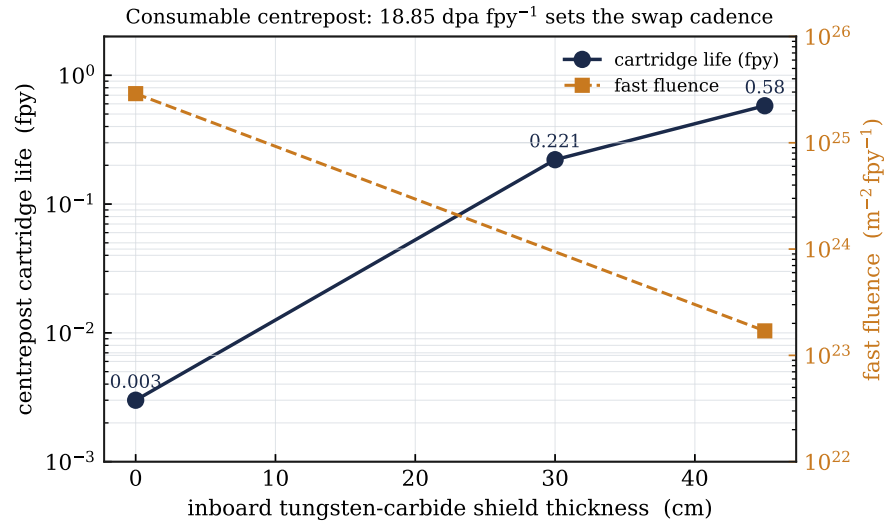


Figure 9. Consumable-centrepost life versus inboard tungsten-carbide shield thickness (left axis) and the corresponding fast-fluence rate (right axis). Anchored at register points BR-L1-A13b (0, 45 cm) and SC-BR-2 (30 cm); the connecting curve is a monotone interpolation of the anchors.

same sites that seed irreversible radiation damage, so for a lifetime-limited magnet the un-doped conductor is preferred — the “un-doped-tape rule” [35]. Both exceed the $3.9 \times 10^{22} \text{ m}^{-2}$ design rule the register upholds, and the first-principles basis of the degradation (chain-oxygen-vacancy formation) is developed in a companion paper [36].

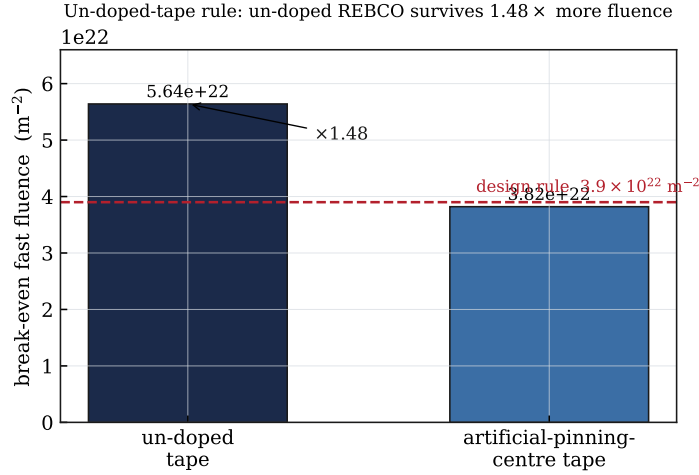


Figure 10. Break-even fast fluence of un-doped versus artificial-pinning-centre REBCO tape (register BR-L2-A36 / HX-22). The un-doped tape tolerates $1.48 \times$ more fluence; both clear the $3.9 \times 10^{22} \text{ m}^{-2}$ design rule.

4.9 Fused quench-precursor detection

The seconds-scale MIITs budget of equation (16) is protected by early precursor detection. Because NI REBCO has a slow v_{NZP} , a developing hot spot can reach a damaging temperature before it produces a measurable resistive voltage, so the twin fuses two non-voltage channels: a fibre-Bragg strain-rate channel reading the mechanical signature of local current redistribution, and a magnetization channel reading the collapse of the screening current. Their quadrature fusion dominates either channel across the receiver-operating characteristic (figure 11, table 1; register KX-L1-A20): area under the curve 0.992 against 0.976 (magnetization) and 0.921 (strain), and — most importantly at the 1% false-positive rate a protection trip demands — a true-positive rate of 0.857 against 0.671 and 0.372. The fusion gain is largest in exactly the low-false-alarm corner where a trip is decided. An in-winding NV-diamond magnetometer provides a local vector precursor at a shot-noise-limited $0.189 \text{ pT Hz}^{-1/2}$ over a 125 kHz bandwidth, with a per-shot signal-to-noise ratio of order 10^9 against the $\sim 100 \text{ mT}$ field step of a shed NI turn (register KX-L1-A22); the detection is developed fully in the companion quench-sensing paper [37].

Table 1. Fused quench-precursor detection (register KX-L1-A20; frozen synthetic, 2×10^4 trials/class). The fused detector dominates the full ROC and is strongest in the low-false-alarm corner.

channel	AUC	TPR @ 1% FPR	TPR @ 0.1% FPR
fibre-Bragg strain-rate	0.921	0.372	0.135
magnetization	0.976	0.671	0.386
fused (strain-rate + magnetization)	0.992	0.857	0.597

4.10 Both magnets qualify

Table 2 collects the result. Under one bore-resolved mechanics benchmarked against the closed form, both magnets qualify structurally, by different margins and against different binding limits. The centrepole is stress-comfortable and neutron-lifetime-limited; the plug is more heavily loaded but still meets its allowable at the required bore. The plug-coil *structural* gate is thereby retired by qualification, not asserted; the residual burner gate lies in the plug-potential physics analysed in the burner papers [30, 31, 32], not in the magnet.

5 Discussion

The results should be read against the published high-field-magnet and REBCO literature.

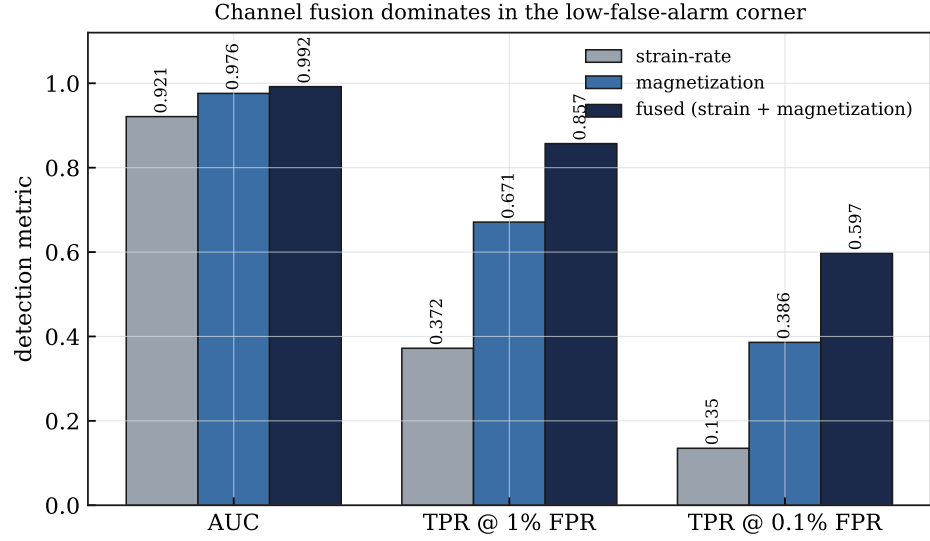


Figure 11. Fused quench-precursor detection metrics (register KX-L1-A20). The fused channel dominates either single channel, with its advantage growing as the false-positive tolerance tightens toward the trip-relevant corner.

Table 2. Both REBCO magnets qualify under one bore-resolved mechanics. Fields are canonical register values; all numbers are design targets or model outputs, not measurements.

quantity	breeder centrepost	burner plug
peak field	16.84 T	26.49 T (design)
field / B_{c2}	0.17	0.26
magnetic pressure	—	115/279/358 MPa (17/26.49/30 T)
reinforcement t_{req}	(stress-comfortable)	0.16/0.48/0.71 m vs 667 MPa
peak hoop strain (graded pre-load)	0.151% CAD	0.283% (0.365% w/ thermal+vertical)
load fraction I_{op}/I_c	0.27	0.43
temperature margin ΔT	49.5 K	37.1 K
binding limit	neutron life (0.58 fpy)	winding strain at required bore
structural verdict	PASS	PASS (FE ratio 0.66)

Field. The field envelope of figure 1 is consistent with the demonstrated record: a REBCO solenoid has reached 45.5 T [3], and a reactor-scale toroidal-field model coil has reached ~ 20 T on the conductor [11, 7]. The Kronos centrepost, at 16.84 T, sits below the demonstrated model-coil field, and the plug design field of 26.49 T sits between the model-coil and solenoid demonstrations — so neither magnet asks the conductor for a field beyond what the literature has shown, and the 35 T conductor-capability point remains a claim about the tape rather than a coil record.

Strain. The 0.40% irreversible-strain ceiling of equation (11) is taken directly from van der Laan’s systematic measurements of the irreversible axial and transverse strain limits of coated conductors [1], and the conductor architecture from the coated-conductor advances reviewed by [2, 6]. The finding that a lifetime-limited winding prefers un-doped tape (figure 10) is consistent with Prokopec’s report that coated-conductor I_c degrades under fast neutron fluence in a manner sensitive to the pinning microstructure [9], and with the fusion-magnet radiation constraints surveyed by [8].

Mechanics. The bore-resolved agreement of finite-element and closed-form Lamé (figure 3) places the reinforcement budget on the standard elasticity of pressurised thick cylinders [25, 26], and the sub-metre reinforcement of figure 6 is within the overwrap and structural practice assumed in conceptual HTS fusion-magnet designs [10, 8]. The no-insulation winding and its self-protection follow the technique introduced by Hahn [5, 4].

Quench. The fused detector (figure 11) should be read against the slow-normal-zone-propagation problem that Marchevsky makes quantitatively [19]: in HTS conductors a hot spot can reach damaging temperature before a resistive voltage appears, so voltage detection alone is inadequate. Our result is complementary to, and evaluated in a harder regime than, the fibre-optic thermometry Salazar demonstrated on VIPER cable at SULTAN [20] and the current-distribution monitoring Teyber showed for fusion cables [21]: the ROC is computed with per-sample signal-to-noise set well below the design-basis fault, and it is precisely in that weak-fault corner that channel fusion recovers early, low-false-alarm warning. The picotesla in-winding NV sensitivity sits comfortably inside the ensemble-NV envelope of the quantum-sensing literature [22, 23, 24].

6 Limitations

Nothing here has been fabricated or measured: every number is a design target, a model output, or an inference from public REBCO literature. Three caveats bound the claims. First, the strain-first envelope of figure 7 is derived from the frozen finite-element anchors through the $p_{\text{mag}} \propto B^2$ scaling with a field-independent thermal/vertical offset; the anchors are finite-element, but the field-swept envelope between them is a model, not a per-field measurement. Second, two conductor elements are reported as open in the register: the graphene-reinforced aluminium bypass is unsupported by data at the operating field, and a specific reinforcement-as-epitaxial-substrate route is infeasible; the structural case does not depend on either. Third, the quench-precursor metrics are twin-synthetic, and the in-winding NV sensor’s neutron tolerance is the one open engineering item, mitigated by shielded siting, a replaceable cartridge and a model-fusion backup, and closed by a dedicated irradiation study. A generic, self-proposed validation roadmap accompanies the analysis in the register [28].

7 Conclusion

The REBCO high-field superconductor is the enabling basis of both Kronos machines, and its binding limit is winding strain, not conductor current. Setting the governing mechanics down as explicit equations — the Lorentz body force, the winding-elasticity boundary-value problem, its closed-form Lamé reduction, the critical-surface margin, the adiabatic quench integral, and the neutron-fluence life law — and reducing them to practice under one bore-resolved finite-element method benchmarked against the closed form, we find that both magnets qualify. The breeder centrepost is stress-comfortable at 16.84 T (load fraction 0.27, $\Delta T \approx 49.5$ K), its binding limit the neutron life of a 0.58-full-power-year scheduled cartridge; the burner plug meets its 667 MPa allowable at its required bore at 26.49 T (load fraction 0.43, $\Delta T \approx 37.1$ K), with a graded per-turn pre-load cutting the peak REBCO hoop strain by 52.4% to hold the winding under the 0.40% irreversible limit. The two magnets are modelled as the distinct devices they are, never conflating 16.84 T with 26.49 T. The value of the work is the strain-first method — the per-turn pre-load and the per-coil, measured-margin qualification — and every number is a design target, not a Kronos measurement.

Acknowledgments

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Reproducibility. Every headline quantity in this paper is a frozen anchor (BU-L1-A2, SC-BR-4, HX-09, HX-10, HX-11, BR-L1-A13b, BR-L2-A36, KX-L1-A15, KX-L1-A20, KX-L1-A22) in the Kronos Physics De-Risking Register [28] (doi:10.5281/zenodo.22133057) and is regenerable from the deposited figure-generation script and the open, CC-BY-4.0 series deposits. This paper is archived at doi:10.5281/zenodo.22645697; official version: <https://www.kronosfusionenergy.com/publications/magnets.html>. No result depends on unpublished or proprietary data, and no economic quantity enters the analysis.

Scope — what this paper does not claim. The 35 T figure is a conductor field-capability claim, not a coil record; the high-field plug-coil feasibility is finite-element and CAD-FEA based, not a built and tested magnet.

Data availability

The field, stress, strain, load-line, neutron-life, irradiation and precursor data used for the figures are archived with the deposit at doi:10.5281/zenodo.22645697 and trace to the register [28] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work — in particular the graded per-turn pre-load winding method — have been filed.

A Derivation of the Lamé reinforcement budget

For a uniform-property reinforced cylinder ($f_r = 0$ in the wall, internal pressure p_{mag} on the bore $r = a$, traction-free outer $r = b$), the homogeneous Navier equation (7) has the general solution $u(r) = C_1 r + C_2/r$. Imposing $\sigma_r(a) = -p_{\text{mag}}$ and $\sigma_r(b) = 0$ through the constitutive law (6) fixes the constants and yields the Lamé stresses

$$\sigma_r(r) = \frac{a^2 p_{\text{mag}}}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right), \quad \sigma_\theta(r) = \frac{a^2 p_{\text{mag}}}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right), \quad (18)$$

the second of which is equation (8). The hoop stress is maximal at $r = a$, giving equation (9); setting $\sigma_\theta(a) = \sigma_{\text{allow}}$ and inverting for b gives the required wall of equation (10). The budget is finite only for $p_{\text{mag}} < \sigma_{\text{allow}}$, satisfied at all three plug fields (figure 5).

B Parameter set

Breeder centrepost: peak field 16.84 T, $B_0 = 8$ T on axis; neutron damage $18.85 \text{ dpa fpy}^{-1}$; cartridge life 0.58 fpy at 45 cm WC; load fraction 0.27; $\Delta T \approx 49.5$ K; CAD peak hoop strain 0.151% ($2.65 \times$ margin). **Burner plug:** design field 26.49 T (evaluation fields 17/26.49/30 T); magnetic pressure 115/279/358 MPa; required reinforcement 0.16/0.48/0.71 m; reinforced allowable 667 MPa; peak hoop strain 0.283% (graded pre-load), 0.365% (with thermal/vertical); load fraction 0.43; $\Delta T \approx 37.1$ K; FE stress ratio 0.66. **Conductor:** irreversible strain limit 0.40%; break-even fast fluence, un-doped 5.64×10^{22} vs pinned $3.82 \times 10^{22} \text{ m}^{-2}$; NV sensitivity $0.189 \text{ pT Hz}^{-1/2}$, bandwidth 125 kHz.

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